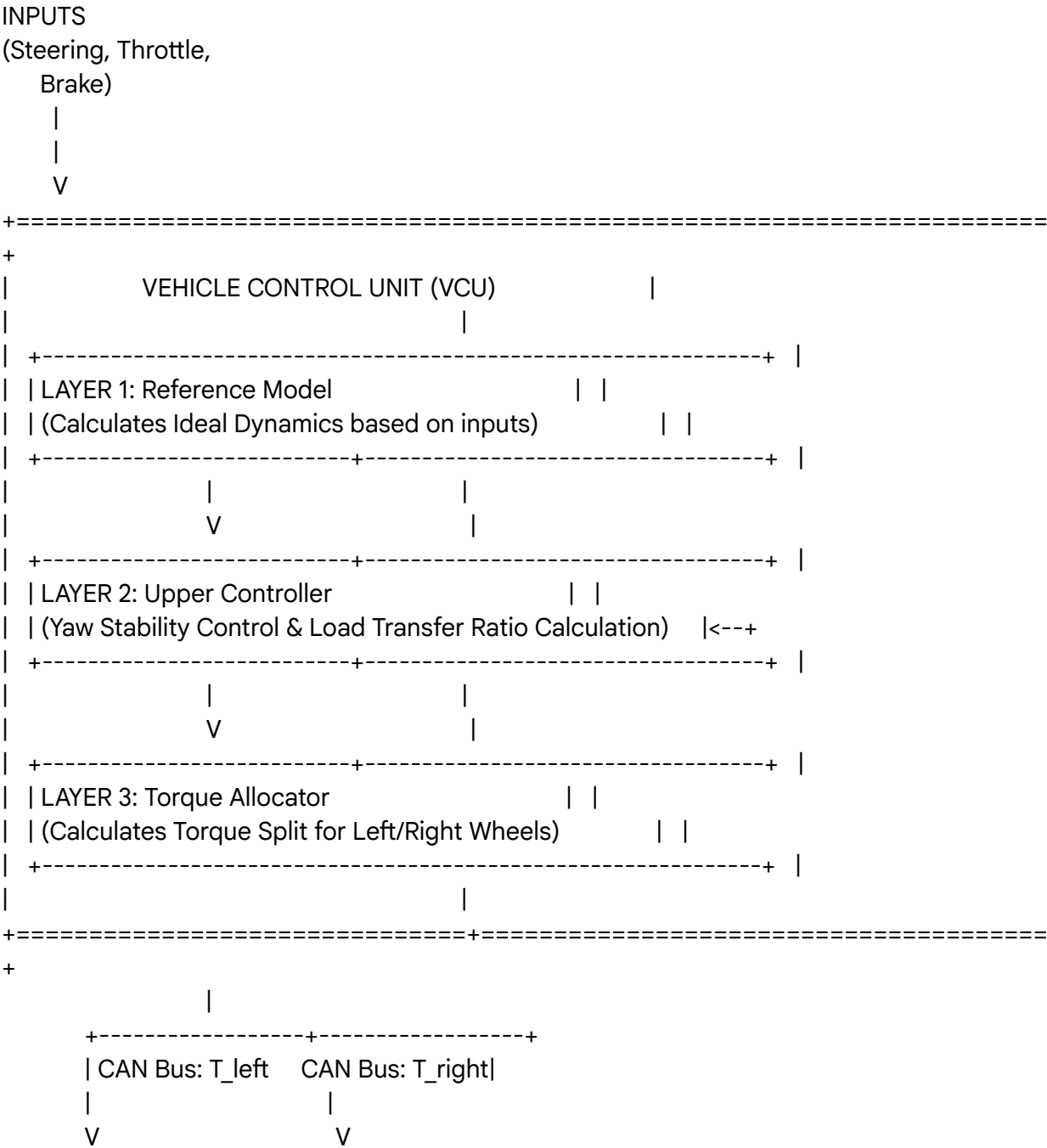
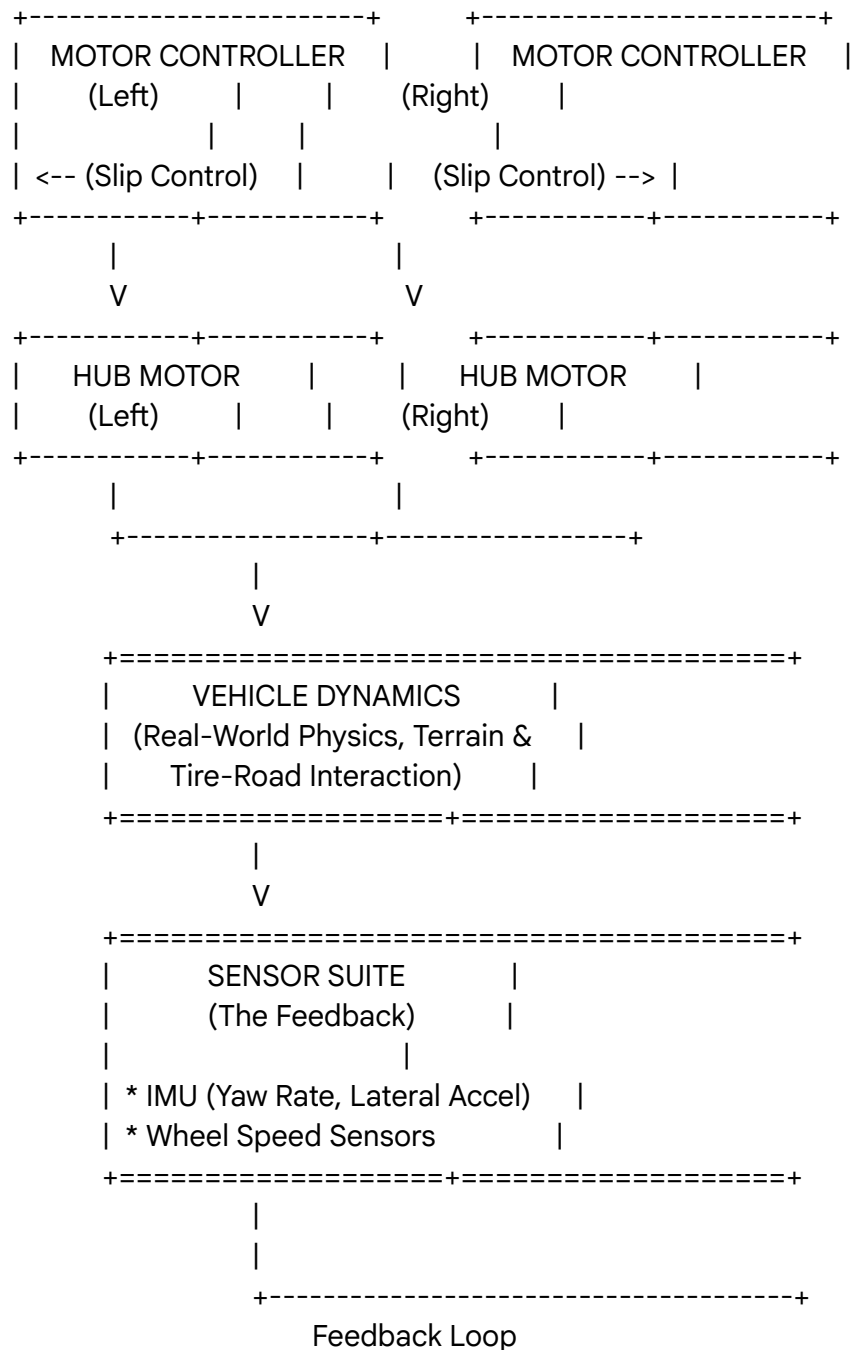


To implement this advanced vehicle-level control system for a 3-wheeler equipped with independent rear hub motors, we must design a system that bridges driver intent, vehicle dynamics, and motor actuation. Based on the constraints of avoiding complex embedded inverter design, this implementation will focus on the **Vehicle Control Unit (VCU)** and its communication with commercial motor controllers.

Here is the detailed block diagram structure and component-level implementation of the control loop.

1. Overall System Block Diagram





2. Detailed Block Implementation & Component Selection

Block A: Sensor Integration (The Feedback Loop)

To execute torque vectoring and rollover prevention, the VCU requires precise, real-time telemetry.

- **Inertial Measurement Unit (IMU):** This is the most critical sensor for the Upper-Level Controller. It measures lateral acceleration (a_y) and yaw rate (ω).

- *Component Choice:* **Bosch BMI088**. This is a high-performance 6-axis automotive-grade IMU specifically designed to suppress harsh vibrations found in robotics and vehicle chassis ``.
- **Wheel Speed Sensors (WSS):** Required at the un-driven front wheel (to determine true vehicle speed) and both rear driven wheels (to calculate slip ratio).
 - *Component Choice:* **Allegro A19352** Giant Magnetoresistance (GMR) sensor ICs mounted near magnetic ring encoders on the wheel hubs, providing extremely accurate speed and directional data ``.
- **Driver Intent Sensors:** A Steering Angle Sensor (SAS) on the handlebars/steering column to measure δ_f , and linear potentiometers on the throttle and brake pedals.

Block B: The Vehicle Control Unit (VCU) - Hierarchical Architecture

The VCU is the "brain" of the system. To handle the mathematical load of first-order physics matrices, Sliding Mode Control (SMC), and Fuzzy Logic without latency, a powerful 32-bit microcontroller is required.

- *Component Choice:* An **STM32** series microcontroller (e.g., STM32F4 or STM32H7). STM32 architectures are highly suitable for implementing discrete digital control systems, Fuzzy Logic controllers, and SMC for vehicle stability in real-time ``.

The software inside the STM32 is partitioned into the three hierarchical layers:

Layer 1: Reference Model

- *Function:* Translates driver inputs into a mathematical ideal. It takes the steering angle (δ_f) and vehicle speed (v_x) and calculates the *desired* yaw rate (ω_{des}) based on a simplified 2-Degree of Freedom (2-DOF) bicycle model. It also translates the throttle position into a total base torque request (T_{base}) ``.

Layer 2: Upper-Level Controller (Stability & Decoupling)

- *Function:* Acts as the dynamic safety net, comparing the driver's ideal path to reality.
- *Implementation:* It calculates the yaw error ($e = \omega_{des} - \omega_{actual}$). Using a Fuzzy Logic or Sliding Mode Controller programmed onto the STM32, it calculates a corrective yaw moment (ΔM_z) required to force the vehicle back onto the intended path ``.
- *Rollover Prevention (LTR):* Simultaneously, this block monitors lateral acceleration (a_y) from the BMI088 IMU. It calculates the Lateral Load Transfer Ratio (LTR). If LTR exceeds a critical threshold (e.g., > 0.8 , indicating the inner wheel is about to lift), the controller enters an override state. It instantly demands a massive negative corrective yaw moment to scrub speed and drop the chassis back down, ignoring the driver's throttle input ``.

Layer 3: Lower-Level Torque Allocator

- *Function:* Translates the physical requirements into specific electrical commands ``.
- *Implementation:* It takes the total desired driving torque and the required corrective yaw moment (ΔM_z) and splits them mathematically.
 - Right Motor Command: $T_{right} = \frac{T_{base}}{2} + \frac{\Delta T}{2}$
 - Left Motor Command: $T_{left} = \frac{T_{base}}{2} - \frac{\Delta T}{2}$

- (Where ΔT is derived from ΔM_z and the vehicle's rear track width).

Block C: Communication Interface

- *Component Choice:* **CAN Bus (Controller Area Network)**.
- *Implementation:* The STM32 VCU packages the calculated T_{left} and T_{right} values and broadcasts them over a CAN 2.0 or CAN FD network. CAN is an automotive standard that utilizes differential signaling to ensure the torque commands reach the motors instantly without being corrupted by the heavy electrical noise generated by the EV powertrain ``.

Block D: Motor Controllers & Torque/Slip Ratio Execution

To adhere to the design constraint of avoiding embedded inverter design, the system utilizes commercial off-the-shelf "smart" motor controllers capable of Field-Oriented Control (FOC).

- *Component Choice:* **VESC** (Vedder Electronic Speed Controller) or **ODrive** hardware . These open-source controllers are gold standards for BLDC motor control, capable of receiving torque commands directly via CAN bus .

Implementing Torque-Based Control & Optimum Slip Ratio:

Instead of the VCU rigidly commanding the wheel to spin at a certain RPM (which causes skipping and dragging), the system utilizes *Torque Control* (Current Control) ``.

1. The VESC receives the targeted torque command (e.g., +50 Nm) via CAN bus ``.
2. The VESC attempts to push exactly that amount of current to the hub motor stators to generate the requested torque ``.
3. *Slip Ratio Enforcement:* The VESC continuously monitors the actual wheel RPM via the motor's internal Hall sensors or the external A19352 sensors. The VCU broadcasts the vehicle's true forward speed (derived from the un-driven front wheel).
4. If the rear wheel's rotational speed drastically exceeds the vehicle's forward speed (indicating a slip ratio $> 15\%$), the tire has broken traction ``. The controller's local anti-slip algorithm instantly overrides the CAN bus torque request, chopping the electrical current to the motor just enough to drop the slip ratio back into the optimal 10-15% window.

This ensures that the wheel is free to rotate at whatever speed the terrain dictates, as long as the tire's friction limit is respected, completely eliminating the mechanical dragging associated with locked axles.